ANY S-MORPHISM BETWEEN SCHEMES WEAKLY ETALE OVER S IS ITSELF WEAKLY ETALE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19164-19168 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19170-19182, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Second card of the weakly etale trilogy; its lemma-key input is the chapter's own and is referenced, not restated.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of weakly etale, base-change and composition stability of flat morphisms (Morphisms, lemmas base-change-flat, composition-flat), and the chapter's key flatness-reflection lemma (lemma-key) used by name.

Source reference. Original lemma 094R: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18919–18923. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01U9: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19182–19182. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01U7: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19182–19182. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of schemes over S. If X, Y are weakly étale over S, then f is weakly étale.*

Proof. We will use Morphisms, Lemmas 01U9 and 01U7 without further mention. Write $X \to Y$ as the composition $X \to X \times_S Y \to Y$. The second morphism is flat as the base change of the flat morphism $X \to S$. The first is the base change of the flat morphism $Y \to Y \times_S Y$ by the morphism $X \times_S Y \to Y \times_S Y$, hence flat. Thus $X \to Y$ is flat. The morphism $X \times_Y X \to X \times_S X$ is an immersion. Thus

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 094R implies, that since X is flat over $X \times_S X$ it follows that X is flat over $X \times_Y X$. $\square$